COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

# STAT 5703

***Statistical Inference and Modeling***

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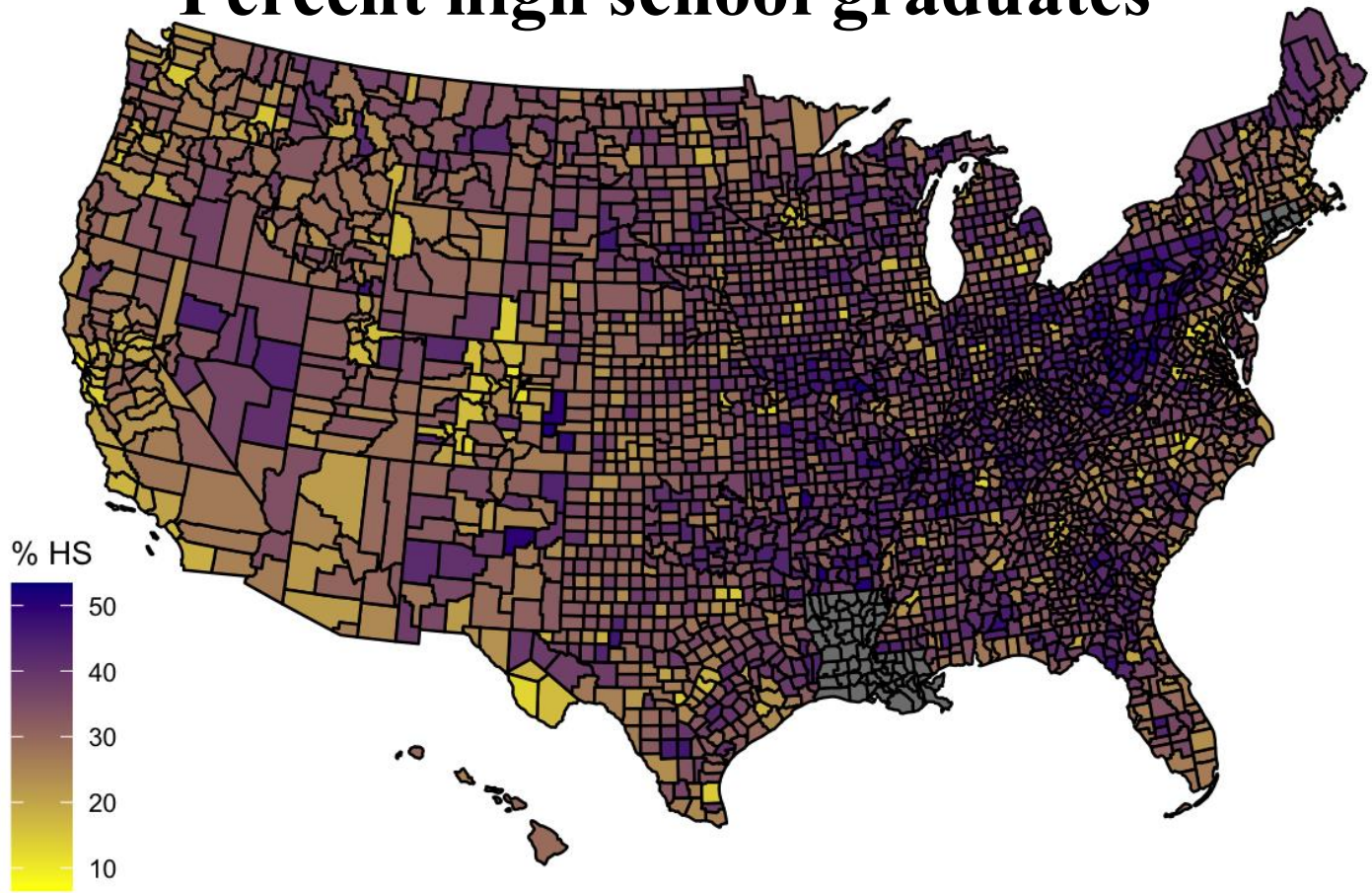
# Motivating Example: Percent high school graduates




- **Research question: what factors contribute to HS graduation rates?**

Data source: U.S. Census Bureau
Time period: 2020 – 2022
Collected by Olivia Huang, Athena Ke, Jaco Hahn

# Hierarchical Data Structure

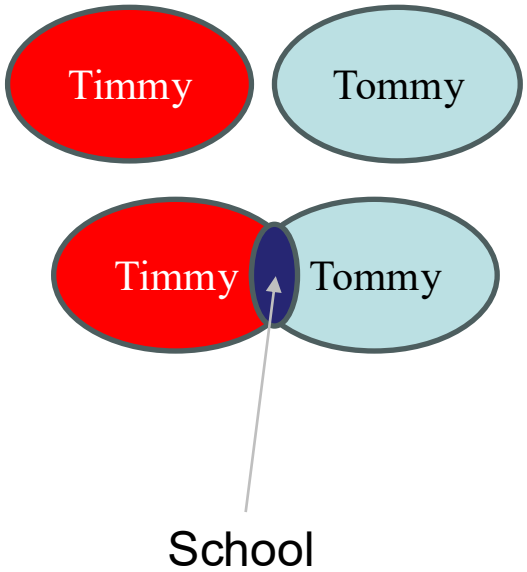
- Suppose we have data on math scores.
- Individual students' scores are nested within schools.
- Schools are nested within counties.
- Counties are nested within states.

Two levels

| State | FIPS | School ID | Student ID | Math score |
|---|---|---|---|---|
| TX | 48465 | 2 | 827 | 63 |
| TX | 48465 | 2 | 640 | 52 |
| TX | 48465 | 96 | 156 | 62 |
| ND | 38091 | 67 | 332 | 14 |
| ND | 38091 | 46 | 343 | 76 |
| OK | 40099 | 41 | 58 | 85 |
| KY | 21091 | 3 | 86 | 37 |

# Hierarchical Data Structure: Two levels

- Suppose we have $m$ groups and data ($\boldsymbol{X}_1, \ldots, \boldsymbol{X}_m$) from each of them, where each $\boldsymbol{X}_j = \left(X_{1j}, \ldots, X_{n_j j}\right)$.
- The random variables, $X_{1j}, \ldots, X_{n_j j}$, nested within Group $j$ *should not be independent*, because they all belong to the same cluster.
- Students within each school perform similarly, because they have the same professors, use same textbooks, etc.
- If I know Timmy's math score and know that Tommy is in the same school as Timmy, I already know a little bit about Tommy's score too.
- The unique contribution of each student is reduced because some information is shared by knowing which school they are in.




*Time to think*: Why do we have to account for clustering?

# Hierarchical Normal Model

- A two-level hierarchical normal model with within group parameters $\left(\theta_j, \sigma^2\right)$ and between group parameter $(\mu, \tau^2)$
- Within group model:
- $X_{1j}, \dots, X_{n_j j} | \theta_j, \sigma^2 \sim N\left(\theta_j, \sigma^2\right), j = 1, \dots, m$
- Between group model:
- $\theta_j | \mu, \tau^2 \sim N(\mu, \tau^2)$

- $\theta_j$ is the average performance of all students within school $j$ (aka random effects), $j = 1, \dots, m$
- $\sigma^2$ measures the within-cluster variability (i.e., at the student level)
- $\mu$ is the average performance of all students in all schools (aka fixed effect)
- $\tau^2$ measures the between-cluster variability (i.e., at school level)

# Hierarchical Data Structure

Consider the normal model for grouped data:

$$X_{1j}, \dots, X_{n_j j} | \theta_j, \sigma^2 \sim N\left(\theta_j, \sigma^2\right), j = 1, \dots, m$$

$$\theta_j | \mu, \tau^2 \sim N(\mu, \tau^2)$$

Let $m = 3$ and $n_1 = n_2 = n_3 = 4$.

It can be shown that $\boldsymbol{X} | \mu, \sigma^2, \tau^2 \sim N_{12}(\mu, \boldsymbol{\Sigma})$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma_1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma_2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Sigma_3} \end{bmatrix}$$

$$\boldsymbol{\Sigma_1} = \boldsymbol{\Sigma_2} = \boldsymbol{\Sigma_3} = \begin{bmatrix} \tau^2 + \sigma^2 & \tau^2 & \tau^2 & \tau^2 \\ \tau^2 & \tau^2 + \sigma^2 & \tau^2 & \tau^2 \\ \tau^2 & \tau^2 & \tau^2 + \sigma^2 & \tau^2 \\ \tau^2 & \tau^2 & \tau^2 & \tau^2 + \sigma^2 \end{bmatrix}$$

The above implies that within cluster correlation is:

$$\rho = \frac{\tau^2}{\tau^2 + \sigma^2}$$

# Hierarchical Data Structure

Consider the normal model for grouped data:

$$\boxed{\begin{gathered} X_{1j}, \dots, X_{n_j j} | \theta_j, \sigma^2 \sim N\left(\theta_j, \sigma^2\right), j = 1, \dots, m \\ \theta_j | \mu, \tau^2 \sim N(\mu, \tau^2) \end{gathered}}$$

Questions:

1. What are the parameters that need to be estimated?
2. How are the parameters estimated?
3. How do we test the main hypothesis of no difference?
4. How do we estimate the group means?

# Hierarchical Data Structure

$$X_{1j}, \dots, X_{n_j j} | \theta_j, \sigma^2 \sim N\left(\theta_j, \sigma^2\right), j = 1, \dots, m$$
$$\theta_j | \mu, \tau^2 \sim N(\mu, \tau^2)$$

1. What are the parameters that need to be estimated?

The parameters, in the sense of frequentist statistics point of view, are only $\mu, \tau^2$ and $\sigma^2$.

Note that this model can easily be extended to regression with mixed effects:

$$y = X\boldsymbol{\beta} + Z_1\theta_1 + \cdots + Z_m\theta_m + \varepsilon, cov(\varepsilon) = \sigma^2 I_n, cov\left(\theta_j\right) = \tau^2 I_{n_j}$$

Then the parameters also include the vector of coefficients $\boldsymbol{\beta}$.

# Hierarchical Data Structure

Consider the normal model for grouped data:

$$X_{1j}, \dots, X_{n_j j} | \theta_j, \sigma^2 \sim N\left(\theta_j, \sigma^2\right), j = 1, \dots, m$$

$$\theta_j | \mu, \tau^2 \sim N(\mu, \tau^2)$$

Questions:

2. How are the parameters $\mu, \tau^2$ and $\sigma^2$ estimated?

If you want to apply the MLE principle, you need to find the marginal likelihood of the data by integrating out all $\theta_j$'s.

That is, the problem of dealing with integrals appears even though this is not a Bayesian model!

# Hierarchical Data Structure

Consider the normal model for grouped data:

$$X_{1j}, \dots, X_{n_j j} | \theta_j, \sigma^2 \sim N\left(\theta_j, \sigma^2\right), j = 1, \dots, m$$
$$\theta_j | \mu, \tau^2 \sim N(\mu, \tau^2)$$

Questions:

3. How do we test the main hypothesis of no difference?

In ANOVA we test $H_0: \theta_1 = \theta_2 = \cdots = \theta_m$

However, this makes no sense when $\theta_j$'s are random variables.

Now we test:

$$H_0: \tau^2 = 0 \; vs. H_a: \tau^2 > 0$$

# Hierarchical Data Structure

Consider the normal model for grouped data:

$$X_{1j}, \dots, X_{n_j j} | \theta_j, \sigma^2 \sim N\left(\theta_j, \sigma^2\right), j = 1, \dots, m$$
$$\theta_j | \mu, \tau^2 \sim N(\mu, \tau^2)$$

Questions:

4. How do we estimate the group means?

This is not even properly phrased. Only fixed parameters can be estimated. In this setup this is called the problem of estimation of realized values of the random components. To differentiate this problem from inference for an estimable parameter, the current problem is often referred to as prediction of a random effect. In general, it is done via $E(\theta_j | \boldsymbol{x})$.

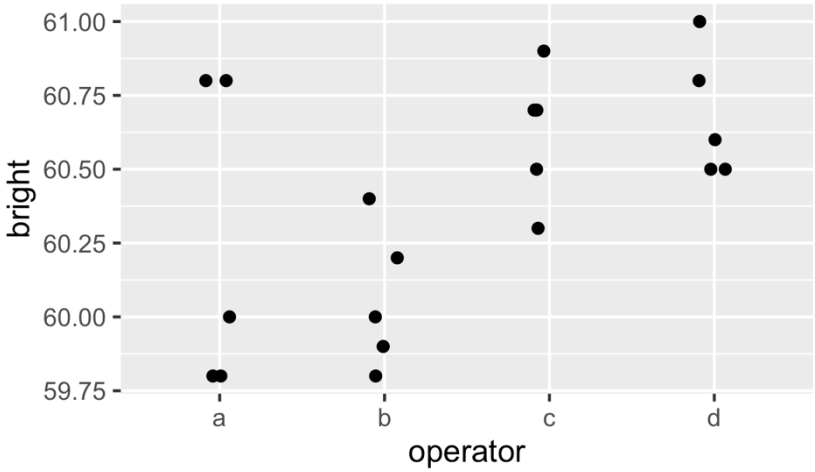
Note: The above issue is automatically solved in a Bayesian model.

# Example: Hierarchical Data Structure

Data on the brightness of paper which may vary between operators of the production machinery.

Questions:

- Is there a difference between operators in general?
- How much is the difference between operators in general?
- How does the variation between operators compare to the variation within operators?
- What is the difference between these four operators?



# Summary: General Hierarchical Data Structure

Two-level hierarchical data model:

$$X_{1j}, \dots, X_{n_j j} | \phi_j \sim f\left(x \middle| \phi_j\right), j = 1, \dots, m$$

$$\phi_1, \dots, \phi_m | \psi \sim iid\ \pi(\phi | \psi)$$

- How should we represent our information about $\phi_1, \dots, \phi_m$?
- Best approach: $\phi_1, \dots, \phi_m$ should be assumed conditionally independent (partial pooling), given a tuning parameter $\psi$.
- The extremes are:
- $\phi_1, \dots, \phi_m$ are "independent" (aka no pooling of information between clusters). Corresponds to $\psi \to \infty$. This is the approach taken by classical ANOVA models.
- $\phi_1, \dots, \phi_m$ are the same (aka complete pooling). Corresponds to $\psi = 0$. In this approach we put all observations together and analyze them as if they came from a single cluster.

# Hierarchical Data Structure

Key question: how should we represent our information about $\phi_1, \ldots, \phi_m$?

The two extremes are:

- $\phi_1, \ldots, \phi_m$ are independent (aka no pooling of information between clusters). This is the approach taken by classical ANOVA models
- $\phi_1, \ldots, \phi_m$ are the same (aka complete pooling). In this approach we put all observations together and analyze them as if they came from a single cluster.
- $\phi_1, \ldots, \phi_m$ are conditionally independent or exchangeable (partial pooling). That is,

$$\phi_1, \ldots, \phi_m | \psi \sim iid\ \pi(\phi|\psi)$$

# Hierarchical Data Structure

Two-level hierarchical data model:

$$X_{1j}, \dots, X_{n_j j} | \phi_j \sim f\left(x \middle| \phi_j\right), j = 1, \dots, m$$

$$\phi_j | \psi \sim \pi(\phi | \psi)$$

- The first line models the *within-group* sampling variability.
- The second line models the *between-group* sampling variability.
- Notice how the parameters are nested within one another.
- Often the parameter $\psi$ will contain some sort of variation component and if it is estimated to be high, this means the clusters were close to independent, and if it is estimated low, this means the clusters were almost identical.
- Let the data decide how much info to pool!

# Issues in Autocorrelated Data

One of the standard assumptions in the regression model is that the error terms $\varepsilon_i$ and $\varepsilon_j$ are uncorrelated for $i \neq j$. When error terms are correlated, problems occur when using OLS estimates:

- Regression Coefficients are still unbiased
- MSE *underestimates* $\sigma^2$
- Standard errors of regression coefficients based on OLS *underestimate* the true standard error
- Inflated $t$ and $F$ statistics and *artificially narrow* confidence intervals

Autocorrelated errors of 1st order:

$$\varepsilon_t = \rho\, \varepsilon_{t-1} + u_t$$

where $u_t$ are uncorrelated disturbances assumed normally distributed.

# First-Order Autoregressive Model

First-Order Autoregressive AR(1) model for simple linear regression:

$$Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t, t = 1, \dots, n, \qquad \varepsilon_t = \rho\varepsilon_{t-1} + u_t$$

The $\rho$ is called autoregressive parameter and we assume that $|\rho| < 1$

The error terms $u_t$ are assumed independent $u_t \sim N(0, \sigma^2)$

Same idea generalizes to multiple regression:

$$\boxed{Y_t = \beta_0 + \beta_1 X_{t1} + \cdots + \beta_p X_{tp} + \varepsilon_t, t = 1, \dots, n, \qquad \varepsilon_t = \rho\varepsilon_{t-1} + u_t}$$

It can be shown that under the assumption the errors are stationary:

$$\varepsilon_t \sim N\left(0, \frac{\sigma^2}{1-\rho^2}\right), cov(\varepsilon_t, \varepsilon_{t-1}) = \frac{\rho\sigma^2}{1-\rho^2}, corr(\varepsilon_t, \varepsilon_{t-1}) = \rho$$

Proof:

$$E(\varepsilon_t) = \rho E(\varepsilon_{t-1}) + E(u_t) = 0$$

$$Var(\varepsilon_t) = Var(\rho\varepsilon_{t-1} + u_t) = \rho^2 Var(\varepsilon_{t-1}) + \sigma^2$$

$$cov(\varepsilon_t, \varepsilon_{t-1}) = cov(\rho\varepsilon_{t-1} + u_t, \varepsilon_{t-1}) = \rho Var(\varepsilon_{t-1}) + cov(u_t, \varepsilon_{t-1})$$

# First-Order Model - II

In General:

$$\varepsilon_t = \rho\varepsilon_{t-1} + u_t = \rho\left(\rho\varepsilon_{t-2} + u_{t-1}\right) + u_t = \rho^2\varepsilon_{t-2} + \rho u_{t-1} + u_t = \ldots = \sum_{s=0}^{\infty}\rho^s u_{t-s}$$

$$E\left\{\varepsilon_t\right\} = 0 \qquad \sigma^2\left\{\varepsilon_t\right\} = \sigma^2\left\{\sum_{s=0}^{\infty}\rho^s u_{t-s}\right\} = \sum_{s=0}^{\infty}\rho^{2s}\sigma^2\left\{u_{t-s}\right\} = \sigma^2\sum_{s=0}^{\infty}\rho^{2s} = \frac{\sigma^2}{1-\rho^2}$$

Covariance: $\sigma\left\{\varepsilon_t, \varepsilon_{t-s}\right\} = \dfrac{\rho^s\sigma^2}{1-\rho^2} \quad s \geq 0$

Correlation: $\rho\left\{\varepsilon_t, \varepsilon_{t-s}\right\} = \rho^s \quad s \geq 0$

$$\sigma^2\left\{\varepsilon\right\} = \frac{\sigma^2}{1-\rho^2}\begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{n-1} \\ \rho & 1 & \rho & \cdots & \rho^{n-2} \\ \rho^2 & \rho & 1 & \cdots & \rho^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \cdots & 1 \end{bmatrix}$$

$AR(2)$: $\quad \varepsilon_t = \rho_1\varepsilon_{t-1} + \rho_2\varepsilon_{t-2} + u_t$ Even Higher or models can be fit as well.

# Recall: Durbin-Watson Test For Independence

Consider the model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i, \qquad \varepsilon_i = \rho\varepsilon_{i-1} + u_i,$$

$$u_i \sim ind. N(0, \sigma^2), |\rho| < 1$$

We would like to test the hypotheses:

$H_0$: $\rho = 0 \Rightarrow$ uncorrelated errors

$H_a$: $\rho > 0 \Rightarrow$ positively correlated errors

Note: The alternative can also be two-sided

Procedure:

1) Fit the regular OLS model and obtain residuals
2) Compute the DW statistic
3) Use p-value (from R) or approximation below

Note: Python does not compute the p-value!

Test statistic:

$$DW = \frac{\sum_{i=2}^{n}(e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2}$$

Note: $E(\varepsilon_i \varepsilon_{i-1}) = \frac{\rho\sigma^2}{1-\rho^2}$

$$\Rightarrow \sum_{i=2}^{n}(e_i - e_{i-1})^2 = \sum_{i=2}^{n} e_i^2 + \sum_{i=2}^{n} e_{i-1}^2 - 2\sum_{i=2}^{n} e_i e_{i-1} \approx 2\sum_{i=1}^{n} e_i^2 - 2n\frac{\rho\sigma^2}{1-\rho^2}$$

Therefore, under $H_0$, we expect DW ≈ 2.

# Autocorrelation - Remedial Measures

- Include a linear time term
- Include seasonal dummy variables if data are quarterly or monthly and residuals show cyclic behavior
- Use transformed Variables that remove the (estimated) autocorrelation parameter (Cochrane-Orcutt and Hildreth-Lu Procedures)
- Use First Differences
- Use Generalized Least Squares which extends the usual linear model to allow for an arbitrary positive definite variance–covariance (or correlation) matrix of $\epsilon$, Var($\boldsymbol{\epsilon}$) = $\boldsymbol{V}\sigma^2$. For example,

$$\boldsymbol{R} = \frac{\sigma^2}{1-\rho^2}\begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n-1} \\ \rho & 1 & \rho & \dots & \rho^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \dots & 1 \end{bmatrix}$$

# Unequal Error Variances

- Here we are discussing the cases $Var(\boldsymbol{\varepsilon}) \neq \sigma^2 \boldsymbol{I}_n$
- Possible reasons:
- Case 1 – Error Variances known exactly (*very* rare case)
- Case 2 – Error Variances known up to a constant
  - Occasionally information known regarding experimental units regarding the relative magnitude (unusual)
  - If "observations" are means of different numbers of units (each with equal variance) at the various *X* levels, Variance of observation *i* is $\sigma^2/n_i$ where $n_i$ is known
- Case 3 – Variance (or Standard Deviation) is related to one or more predictors, and relation can be modeled (like Breusch-Pagan Test)
- This usually results in biased inconsistent underestimated SEs.
- Ordinary Least Squares must be performed with correct variances

# Case 1 Known Variances - I

Consider the multiple regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i, \varepsilon_i \sim N\left(0, \sigma_i^2\right), i = 1, \ldots, n, cov\left(\varepsilon_i, \varepsilon_j\right) = 0, \forall i \neq j$$

$$\Rightarrow cov(\boldsymbol{\varepsilon}) = \begin{bmatrix} \sigma_1^2 & 0 & \ldots & 0 \\ 0 & \sigma_2^2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \sigma_n^2 \end{bmatrix}$$

Then using the maximum likelihood method, we must maximize

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \left[ \frac{1}{\sqrt{2\pi\sigma_i^2}} e^{-\frac{1}{2\sigma_i^2}\left(Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_p X_{ip}\right)^2} \right]$$

Set $w_i = \frac{1}{\sigma_i^2}$ and rewrite the likelihood as:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \left[ \sqrt{\frac{w_i}{2\pi}} \right] e^{-\frac{1}{2}\sum_{i=1}^{n} w_i\left(Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_p X_{ip}\right)^2}$$

That is, to maximize $L(\boldsymbol{\beta})$ we need to minimize

$$Q_w(\beta) = \sum_{i=1}^{n} w_i\left(Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_p X_{ip}\right)^2$$

Which is a weighted version of the regular SSE.

# WLS – Case 1 Known Variances - II

Easiest to set up in matrix form, where: $\mathbf{W} = \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_n \end{bmatrix} = \mathbf{W}'$

$$\sigma^2\left\{\mathbf{Y}\right\} = \sigma^2\left\{\varepsilon\right\} = \mathbf{W}^{-1}$$

Normal Equations: $\left(\mathbf{X'WX}\right)\mathbf{b}_w = \mathbf{X'WY}$

$$\Rightarrow \quad \mathbf{b}_w = \left(\mathbf{X'WX}\right)^{-1}\mathbf{X'WY} = \mathbf{AY} \quad \mathbf{A} = \left(\mathbf{X'WX}\right)^{-1}\mathbf{X'W}$$

$$\Rightarrow \quad \mathbf{E}\left\{\mathbf{b_w}\right\} = \mathbf{AE}\left\{\mathbf{Y}\right\} = \mathbf{AX}\beta = \left(\mathbf{X'WX}\right)^{-1}\mathbf{X'WX}\beta = \beta$$

$$\Rightarrow \quad \sigma^2\left\{\mathbf{b_w}\right\} = \mathbf{A}\sigma^2\left\{\mathbf{Y}\right\}\mathbf{A'} = \left(\mathbf{X'WX}\right)^{-1}\mathbf{X'WW^{-1}WX}\left(\mathbf{X'WX}\right)^{-1} = \left(\mathbf{X'WX}\right)^{-1}$$

# WLS – Case 2 – Variance Known up to Constant - I

Data are means of unequal numbers of replicates at X-levels, can use any weights generally

$$\sigma^2\left\{\overline{Y}_i\right\} = \sigma_i^2 = \frac{\sigma^2}{r_i} \quad \Rightarrow \quad \sigma^2\left\{\mathbf{Y}\right\} = \sigma^2 \begin{bmatrix} r_1^{-1} & 0 & \cdots & 0 \\ 0 & r_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_n^{-1} \end{bmatrix} \qquad r_i \equiv \text{ number of replicates at } i^{th} \text{ level of } X$$

$$w_i = \frac{1}{\sigma_i^2} = \frac{n_i}{\sigma^2} \quad \Rightarrow \quad \mathbf{W} = \frac{1}{\sigma^2}\begin{bmatrix} r_1 & 0 & \cdots & 0 \\ 0 & r_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_n \end{bmatrix} = \frac{1}{\sigma^2}\mathbf{W}^* \qquad \mathbf{W}^* = \begin{bmatrix} r_1 & 0 & \cdots & 0 \\ 0 & r_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_n \end{bmatrix}$$

$$\Rightarrow \mathbf{X'WX} = \mathbf{X'}\frac{1}{\sigma^2}\mathbf{W}^*\mathbf{X} = \frac{1}{\sigma^2}\mathbf{X'W}^*\mathbf{X} \Rightarrow \left(\mathbf{X'WX}\right)^{-1} = \sigma^2\left(\mathbf{X'W}^*\mathbf{X}\right)^{-1}$$

$$\Rightarrow \mathbf{X'WY} = \mathbf{X'}\frac{1}{\sigma^2}\mathbf{W}^*\mathbf{Y} = \frac{1}{\sigma^2}\mathbf{X'W}^*\mathbf{Y} \quad \Rightarrow \mathbf{b}_w = \left(\mathbf{X'WX}\right)^{-1}\mathbf{X'WY} = \left(\mathbf{X'W}^*\mathbf{X}\right)^{-1}\mathbf{X'W}^*\mathbf{Y}$$

$$\sigma^2\left\{\mathbf{b}_w\right\} = \left(\mathbf{X'WX}\right)^{-1} = \sigma^2\left(\mathbf{X'W}^*\mathbf{X}\right)^{-1} \qquad \mathbf{s}^2\left\{\mathbf{b}_w\right\} = MSE_w\left(\mathbf{X'W}^*\mathbf{X}\right)^{-1} \qquad MSE_w = \frac{\sum_{i=1}^{n} w_i^*\left(Y_i - \hat{Y}_i\right)^2}{\left(n-p\right)}$$

# WLS - Case 3 – Estimated Variances

Use squared residuals (estimated variances) or absolute residuals (estimated standard deviations) from OLS to model their levels as functions of predictor variables

- Plot Residuals, squared residuals and absolute residuals versus fitted values and predictors
- Using model building techniques used on *Y* previously to obtain a model for either the variances or standard deviations as functions of the *X*'s or the mean
- Fit estimated WLS with the estimated weights (1/variances)
- Iterate until regression coefficients are stable (iteratively re-weighted least squares):

$w_i = \frac{1}{\hat{s}_i^2}$, where $\hat{s}_i$ is the fitted value from the residual regression

$$b_w = (X'WX)^{-1}X'WY$$

# OLS with Adjusted Variances

$$\mathbf{E}\{\mathbf{Y}\} = \mathbf{X\beta} \qquad \boldsymbol{\sigma}^2\{\mathbf{Y}\} = \boldsymbol{\sigma}^2\{\boldsymbol{\varepsilon}\} = \begin{bmatrix} \sigma_1^2 & 0 & \mathrm{L} & 0 \\ 0 & \sigma_2^2 & \mathrm{L} & 0 \\ \mathrm{M} & \mathrm{M} & \mathrm{O} & \mathrm{M} \\ 0 & 0 & \mathrm{L} & \sigma_n^2 \end{bmatrix}$$

$$\mathbf{b} = (\mathbf{X'X})^{-1}\mathbf{X'Y} = \mathbf{AY} \qquad \mathbf{A} = (\mathbf{X'X})^{-1}\mathbf{X'}$$

$$\Rightarrow \quad \boldsymbol{\sigma}^2\{\mathbf{b}\} = \mathbf{A}\boldsymbol{\sigma}^2\{\mathbf{Y}\}\mathbf{A'} = (\mathbf{X'X})^{-1}\mathbf{X'}\boldsymbol{\sigma}^2\{\boldsymbol{\varepsilon}\}\mathbf{X}(\mathbf{X'X})^{-1}$$

Note: $E\{\varepsilon_i^2\} = \sigma_i^2$ Use $e_i^2$ as an estimator of $\sigma_i^2$ : $\mathbf{S_0} = \begin{bmatrix} e_1^2 & 0 & \mathrm{L} & 0 \\ 0 & e_2^2 & \mathrm{L} & 0 \\ \mathrm{M} & \mathrm{M} & \mathrm{O} & \mathrm{M} \\ 0 & 0 & \mathrm{L} & e_n^2 \end{bmatrix}$

$$\mathbf{s}^2\{\mathbf{b}\} = (\mathbf{X'X})^{-1}\mathbf{X'S_0X}(\mathbf{X'X})^{-1}$$

This is referred to White's estimator, and is robust to specification of form of error variance